# Celal Tarık Bayraktar

Software Developer · C#/ASP.NET Core · React · SQL

Ankara, Türkiye · ctbayraktar6809@gmail.com · Military service: deferred until 05/2029
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## SUMMARY

Software developer and electrical-electronics engineer, building full-stack web applications in C#/ASP.NET Core, React and SQL at Türkiye's Ministry of Trade, and supporting its customs transit (NCTS) data operations in Oracle. Outside work, ships products end to end by directing Claude Code: a web-based SPICE circuit simulator, two live iPhone games and a Unity simulation in development. Data Science MSc student at Gazi University.

## EXPERIENCE

**Software Developer (Assistant Trade Expert)** Aug 2025 – present

Ministry of Trade, General Directorate of Information Technologies (BTGM) · Ankara

- Modernize the ministry's ETAP and TEKİZ web platforms with React, C#/ASP.NET Core Web API and SQL.
- Build React screens and components, and the ASP.NET Core services, controllers and business logic behind them.
- Design new SQL tables and reshape existing schemas for the redesigned workflows.
- Diagnose and fix production defects, and improve existing features.
- Support customs transit operations under NCTS: write Oracle queries in Toad to trace status and data-consistency problems, and resolve records that stall in the expected flow, working with the General Directorate of Customs.

**Engineering Intern** Aug – Sep 2022

Koçak Mühendislik · Aksaray

- Supported the engineering team with calculations, technical drawings and site visits.

## PROJECTS

**PlainSpice** · Circuit simulator for the web · plainspice.com 2026

- Beginner-first circuit simulator for students and my Solved Circuits channel: ngspice 47 compiled to WebAssembly, matching desktop ngspice to six significant figures, with the results written on the drawing.
- React 19 and TypeScript: schematic editor synced with the netlist, plain-language checks for beginner mistakes, share links, offline use; WCAG 2.2 AA, Playwright and axe tests, CI to GitHub Pages.

**One More Passenger** · iPhone game · live on the App Store 2026

- Physics stacking game: drop passengers onto a rocking ferry sailing from Istanbul to Bodrum.
- HTML5 canvas game in an Expo (React Native) app, built in the cloud without a Mac; English and Turkish; AdMob ads with Google's consent flow; 254 end-to-end tests and bot playtests for balance.

**Don't Spill: Coffee Delivery** · iPhone game · live on the App Store 2026

- One-touch physics game built with Capacitor; launched October 2026, v1.1 two weeks later; in-app purchases.

**From Flip-Flops to Atari** · Unity (C#) simulation game In development

- Educational game: build a computer from logic gates up to a working console, packaging each circuit as a reusable module; runs on a pure-C# deterministic simulation engine kept separate from Unity.

**Educational YouTube channels** · Automated production pipelines 2026

- Two channels (computer internals, worked circuit-analysis problems) whose scripting, narration, rendering and publishing run as scheduled jobs.

**NLP e-commerce chatbot** · Python · Flask · scikit-learn · spaCy · NLTK

- Chatbot for an online clothing store: Naive Bayes intent classification, plus two spaCy NER models and rule-based color detection to extract product, size and color.

## SKILLS

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|---|---|
| **Languages** | C#, JavaScript, Python, SQL, C++, MATLAB |
| **Backend** | ASP.NET Core Web API, .NET, REST APIs |
| **Frontend and mobile** | React, HTML/CSS, Capacitor, Expo, iOS App Store publishing (TestFlight, in-app purchases) |
| **Data** | Oracle (Toad), SQL schema design, pandas, scikit-learn, spaCy, NLTK |
| **Game development** | Unity (C#) |
| **Tools** | Git/GitHub, Visual Studio, VS Code, AI-assisted development (Claude Code) |

## EDUCATION

**MSc, Data Science** · Gazi University Feb 2026 – present

**BSc, Electrical-Electronics Engineering** · Ankara University 2023

## LANGUAGES

**English** Advanced (YDS 93.75/100, 2024)

**Turkish** Native